

ISEN/DAEN 427 Lecture 7

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An Introduction to Linear Regression
in progress, new slides will be included

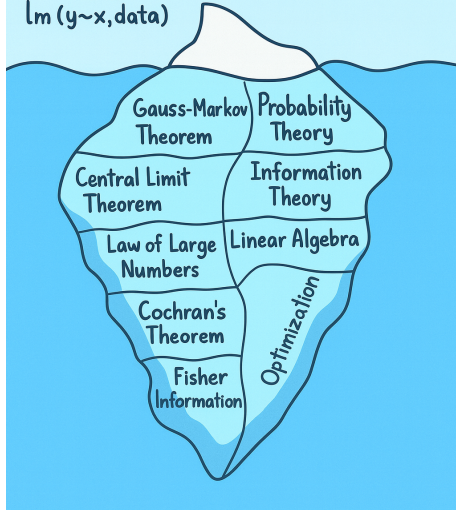


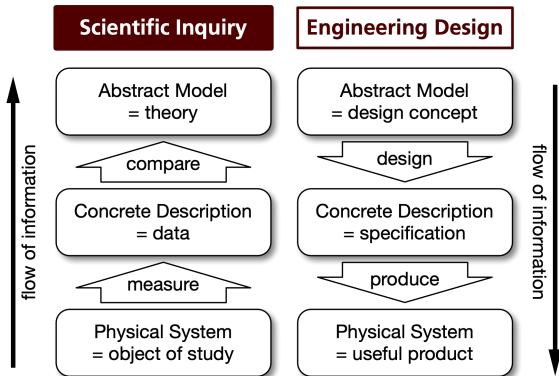


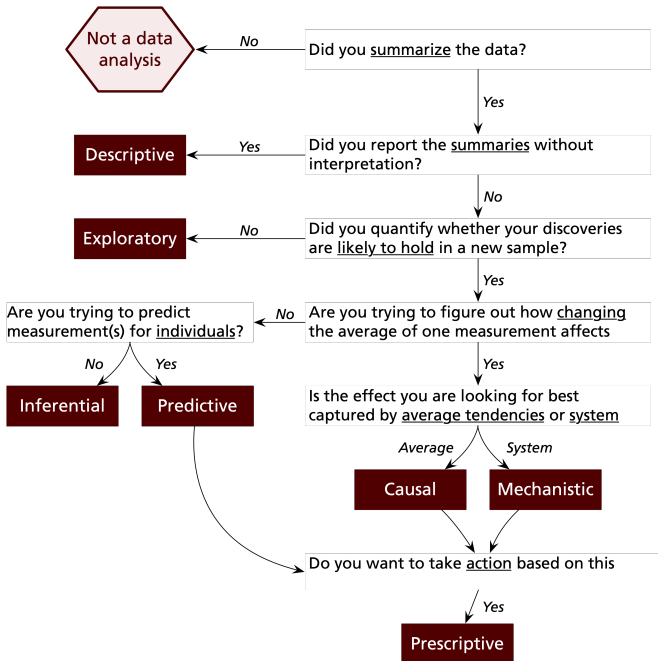
`lm(y~x,data)`

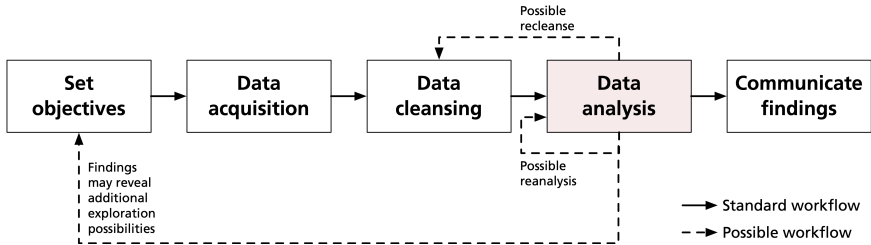


`LinearRegression().fit(x, y)`









Learning Goals

- Understand what a linear regression model is.
- Interpret the meaning of a line of best fit.
- Use a simple linear equation to make predictions.

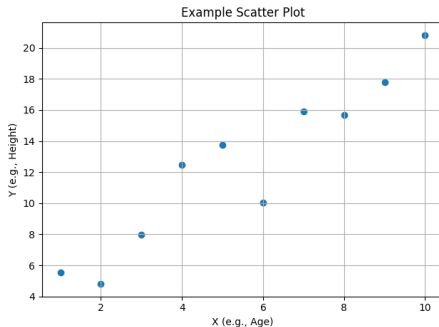
What is Linear Regression?

Definition

Linear regression is a way to describe the relationship between two variables using a straight line.

Example

You might want to predict someone's height based on their age.



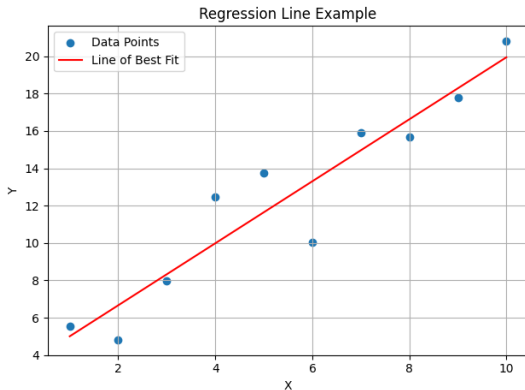
The Equation of a Line

The line of best fit can be written as:

$$\hat{y} = b_0 + b_1x$$

- $\hat{y}$ = predicted value of y
- x = independent variable (what we know)
- b_0 = intercept (value of y when $x = 0$)
- b_1 = slope (change in y for each 1-unit increase in x)

Visualizing the Line of Best Fit



- The line passes close to most of the data points.
- It shows the general trend — not a perfect fit for all data.

Interpreting the Slope and Intercept

Slope b_1

Tells us how much y changes when x increases by 1.

Intercept b_0

Tells us the predicted value of y when $x = 0$.

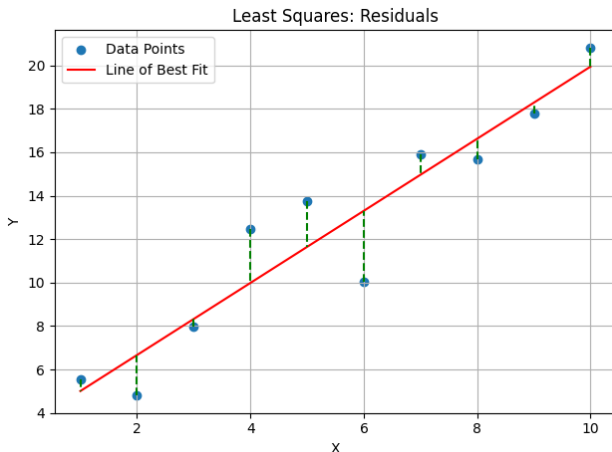
Example

If $\hat{y} = 2 + 3x$:

- The slope is 3: for every increase of 1 in x , y increases by 3.
- The intercept is 2: when $x = 0$, $y = 2$.

How Do We Find the Line?

- We choose the line that minimizes the total distance between the line and all data points.
- This is called the **least squares method**.



Using the Model for Prediction

Example

If our model is:

$$\hat{y} = 50 + 5x$$

and $x = 4$, then:

$$\hat{y} = 50 + 5(4) = 70$$

- We can use the regression line to predict future or missing values.

Linear regression works best when:

- The relationship between x and y is approximately linear.
 - There are no extreme outliers.
 - The data points are evenly spread along the line.
-
- But ... what about binary variables? (*next*)

Exercise

Given the data below, draw a rough line of best fit:

$$(1, 2), (2, 4), (3, 5), (4, 4), (5, 6)$$

Then, estimate the slope and intercept.

Code

```
from sklearn.linear_model import LinearRegression
import numpy as np
import matplotlib.pyplot as plt

# example data
X = np.array([[1], [2], [3], [4], [5]])
y = np.array([2, 4, 5, 4, 6])

model = LinearRegression().fit(X, y) # fit model

print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)

# predict
X_new = np.array([[6], [7]])
print("Predictions:", model.predict(X_new))

plt.figure(figsize=(5, 4))
plt.scatter(X, y)
plt.plot(X, model.coef_[0]*X + model.intercept_, color="red")
plt.grid(True)
```

Code for next class

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import arviz as az
import pymc as pm
from scipy import stats

# real data
penguins = pd.read_csv('/Users/abuabara/TAMU/Teaching/2025F/427/data/penguins.csv')
missing_data = penguins[[
    'body_mass_g', 'flipper_length_mm', 'bill_length_mm', 'sex'
]].isnull().any(axis=1)
penguins = penguins.loc[~missing_data]
penguins.head()

counts = penguins['sex'].value_counts().sort_index()
counts
```

Code for next class

```
plt.figure(figsize=(5, 4))
counts.plot(kind='bar')
plt.xticks([0, 1], ['Male (0)', 'Female (1)'], rotation=0)
plt.xlabel('Sex')
plt.ylabel('Count')
# plt.title('Distribution')
plt.show()
```
